

Domain 1 — Tyre Degradation

From analysis → modeling (generated 2026-01-07)

This report summarizes an end-to-end pipeline for predicting tyre degradation in Formula 1.

It starts from data ingestion and exploratory analysis, then moves through feature engineering and and finishes with diagnostics, limitations, and operational guidance.

- Data: FastF1-derived stints (train: 2022-2023, test: 2024)
- Target: degradation rate (ms/lap)
- Model: XGBoost regression + quantile models for uncertainty
- Holdout performance: $R^2 = 0.65$, $MAE \approx 103$ ms/lap
- Key features: shape_cluster, teammate_delta, circuit_cluster

How to read the charts:

- Predicted vs Actual: points near the diagonal are accurate
- Residuals: centered around 0 means low bias; wider spread means uncertainty
- Segment views (circuit/driver/compound): reveal where the model works and fails

Note: Wet conditions (INTERMEDIATE/WET) are underrepresented and behave differently.

Treat wet-tyre predictions as out-of-scope for this first version.

Executive Summary

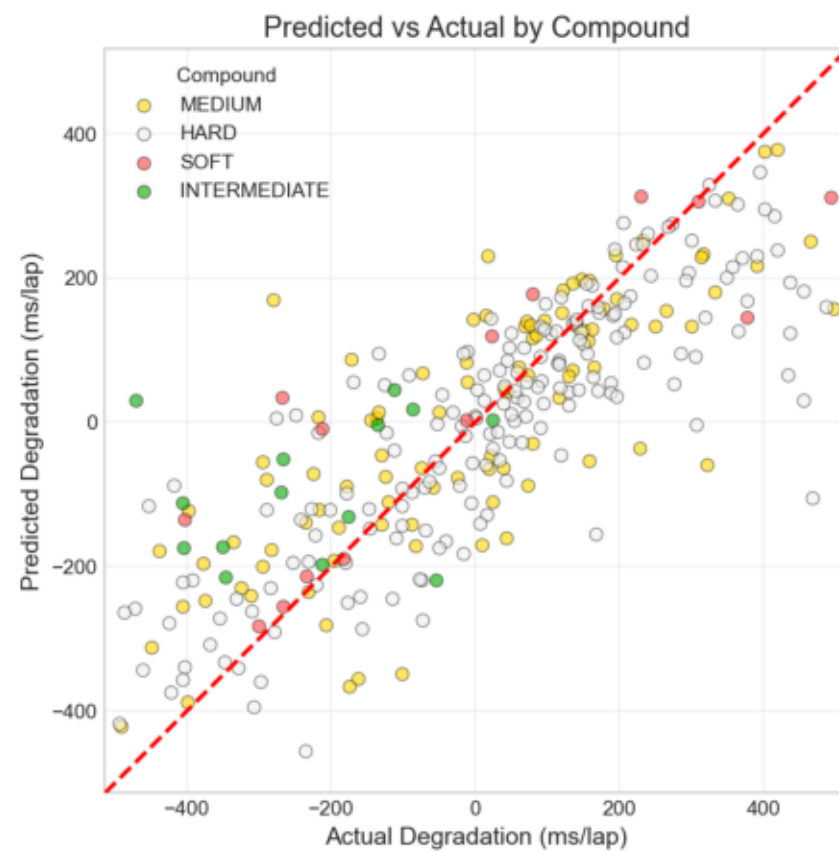
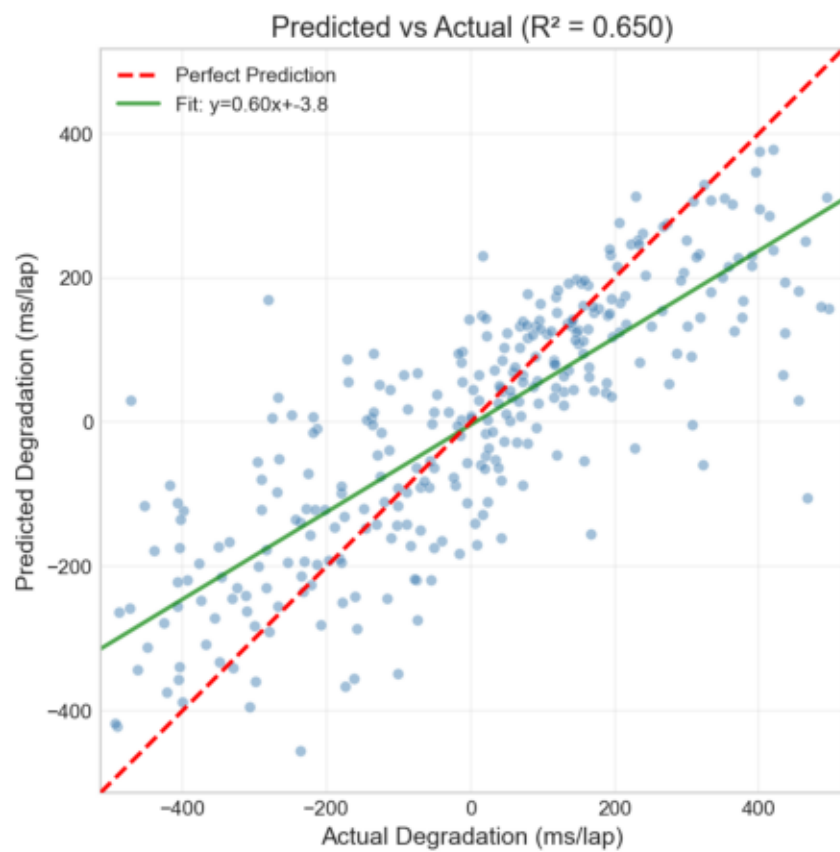
What we learned:

- Degradation is best predicted by **shape** (curve behavior) and **relative driver effect** (team/ma
- Circuit identity matters, but less than the degradation regime and driver-in-car signal.
- Errors are small on typical dry stints, but grow at extremes and in wet conditions.
- Some circuits show systematic under/over-prediction (surface + setup + season drift).

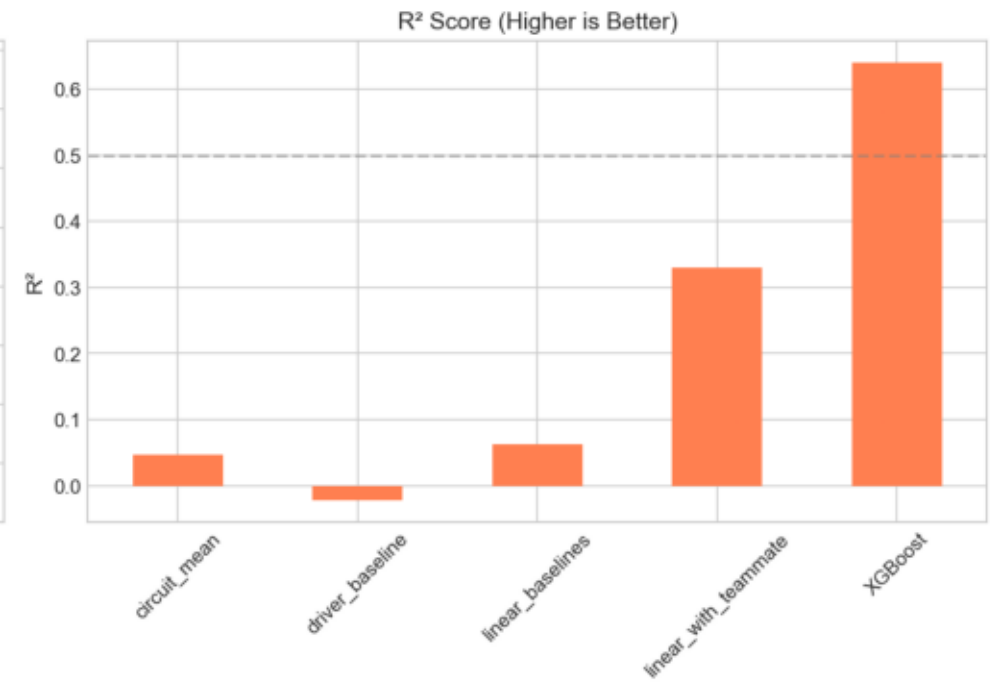
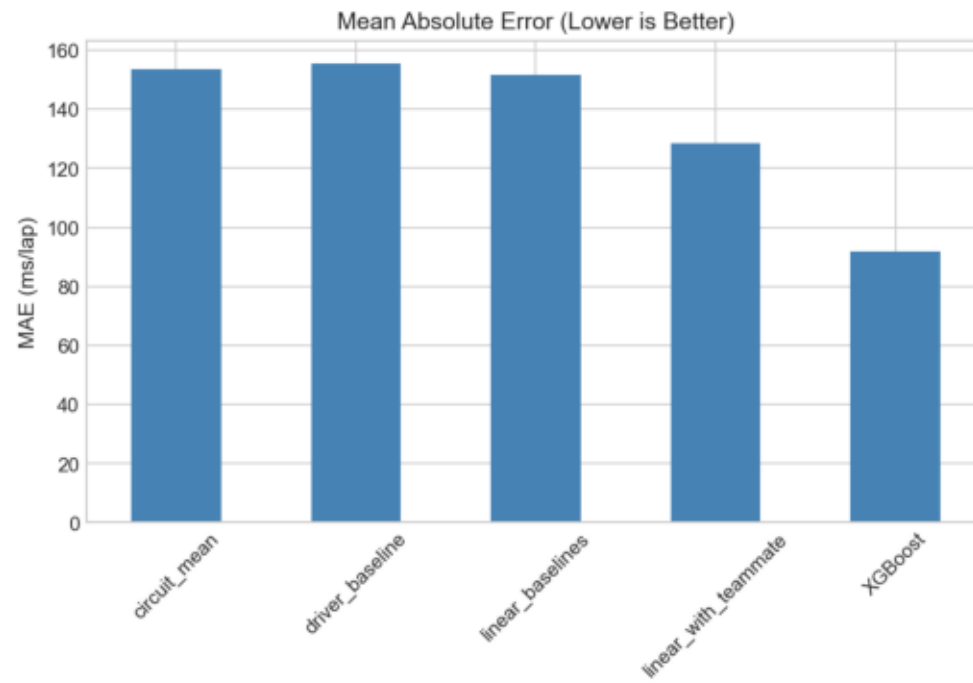
Operational guidance:

- Use model outputs to bound pit windows and compare strategies (especially for dry races).
- Apply circuit-specific monitoring on tracks with higher error (e.g., high-variance weekends).
- For safety, widen prediction intervals (quantiles under-cover in early calibration).
- Build a dedicated wet model (or fallback rules) before using the model in rain.

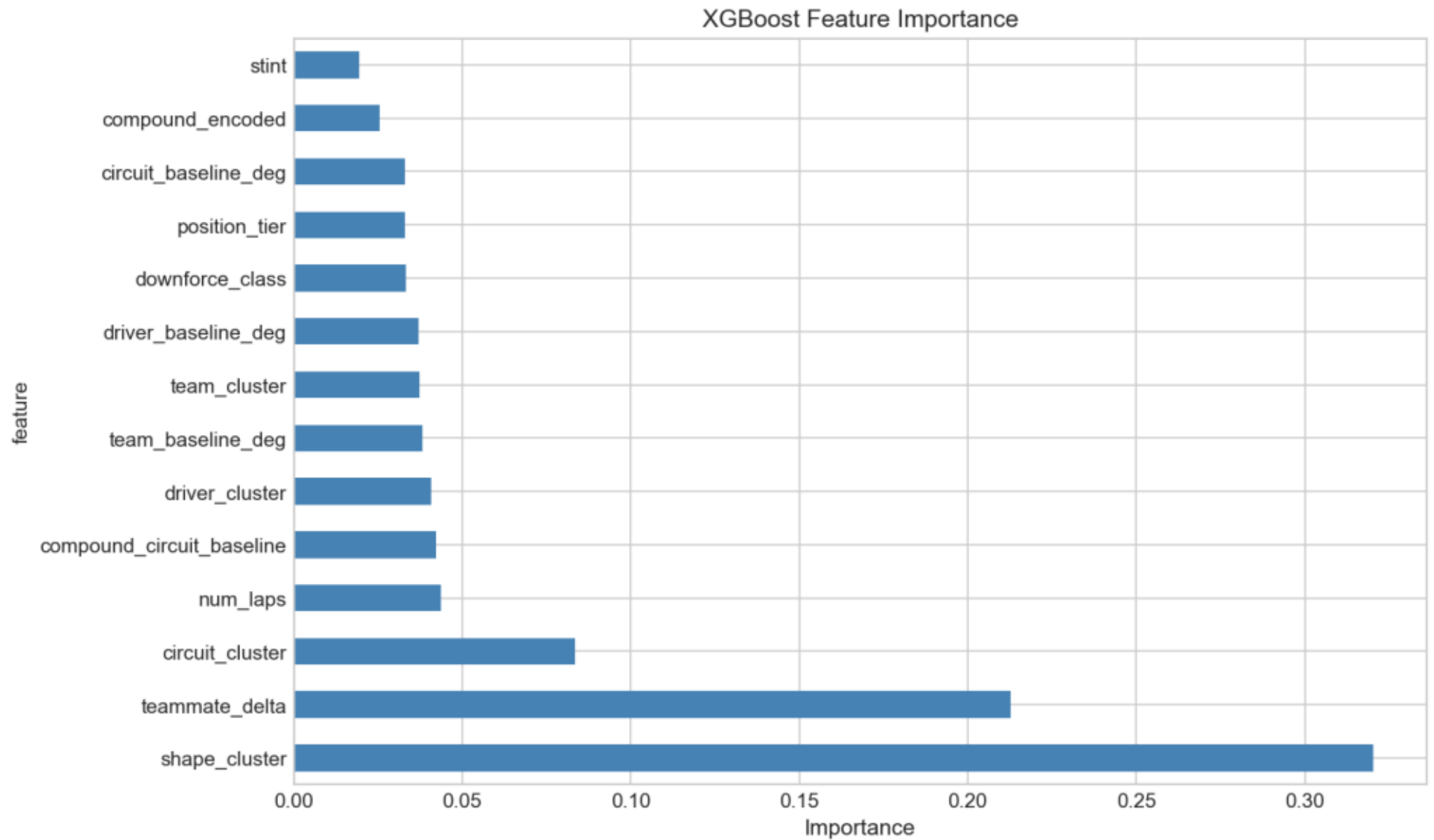
Predicted vs Actual (overall + by compound)



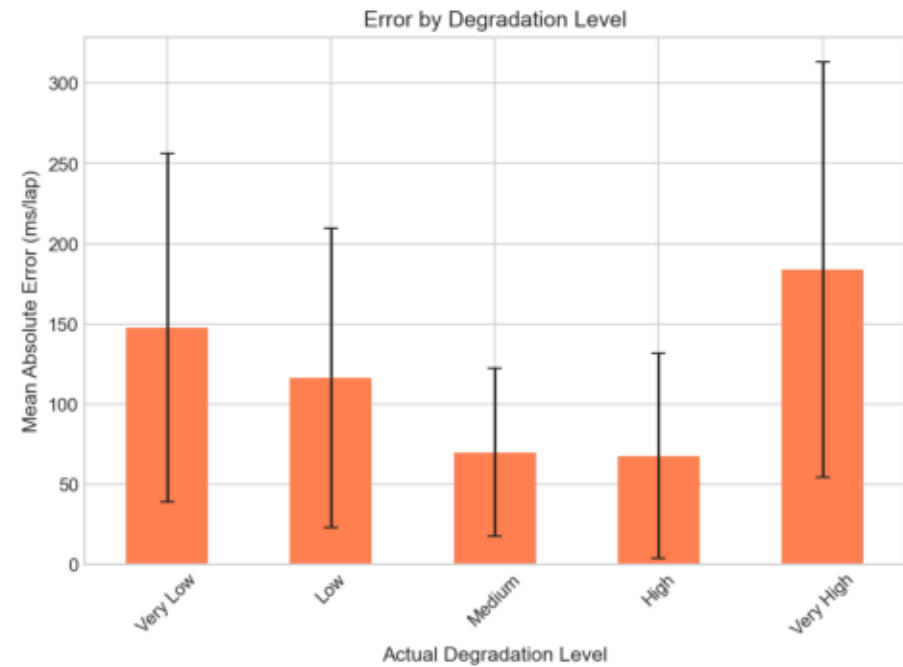
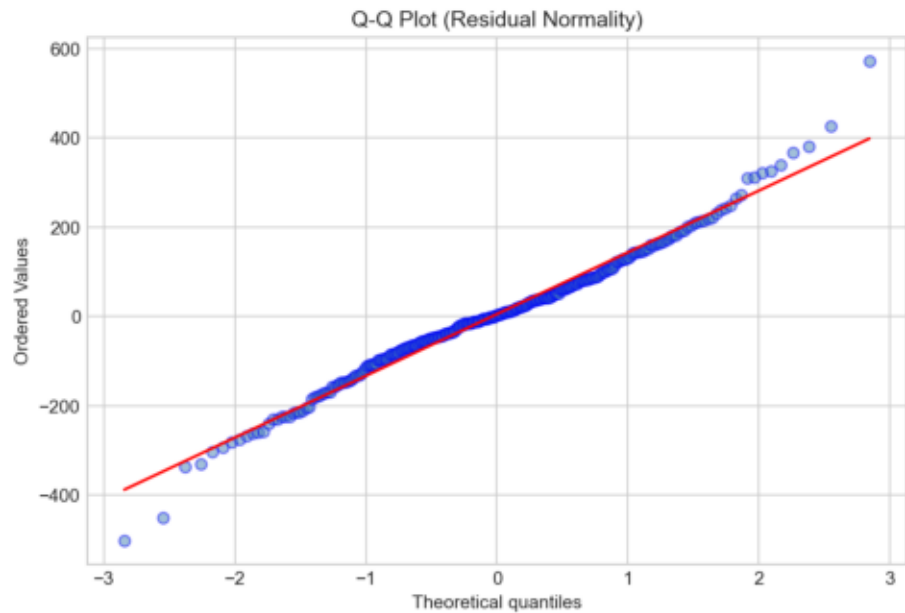
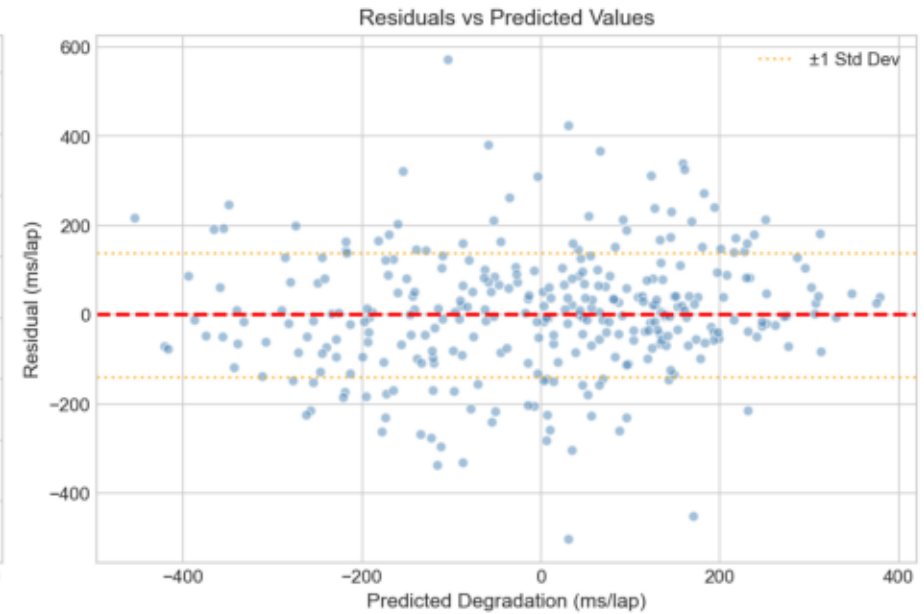
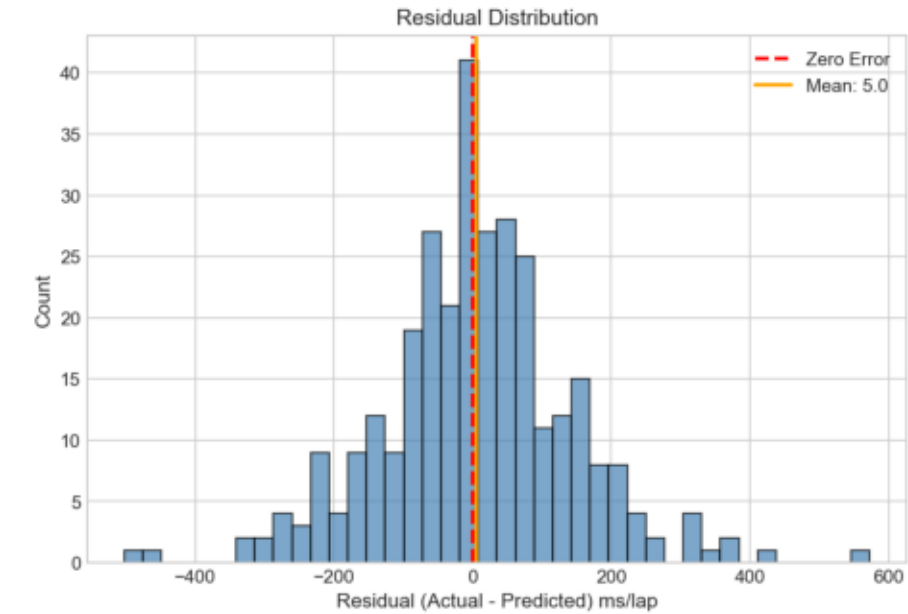
Baseline comparison vs XGBoost



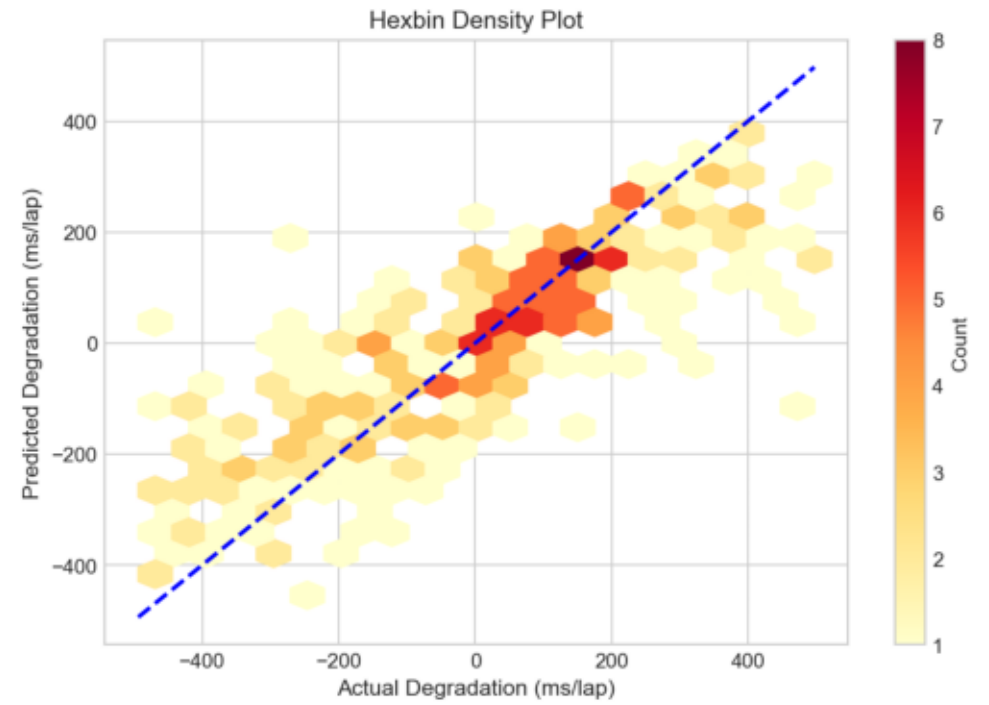
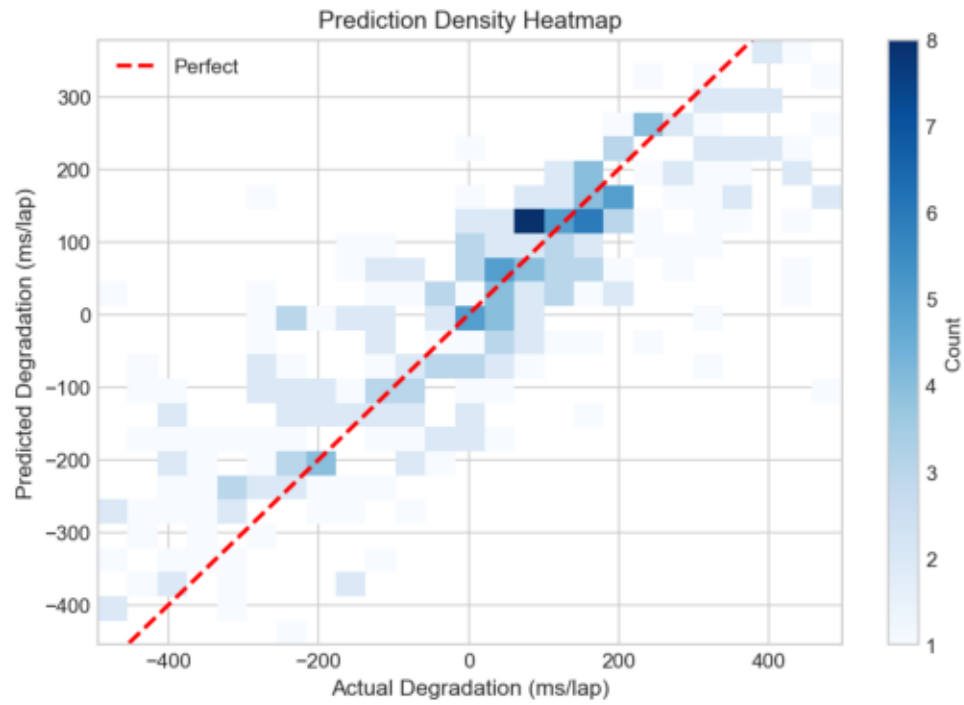
Feature importance (top drivers of performance)



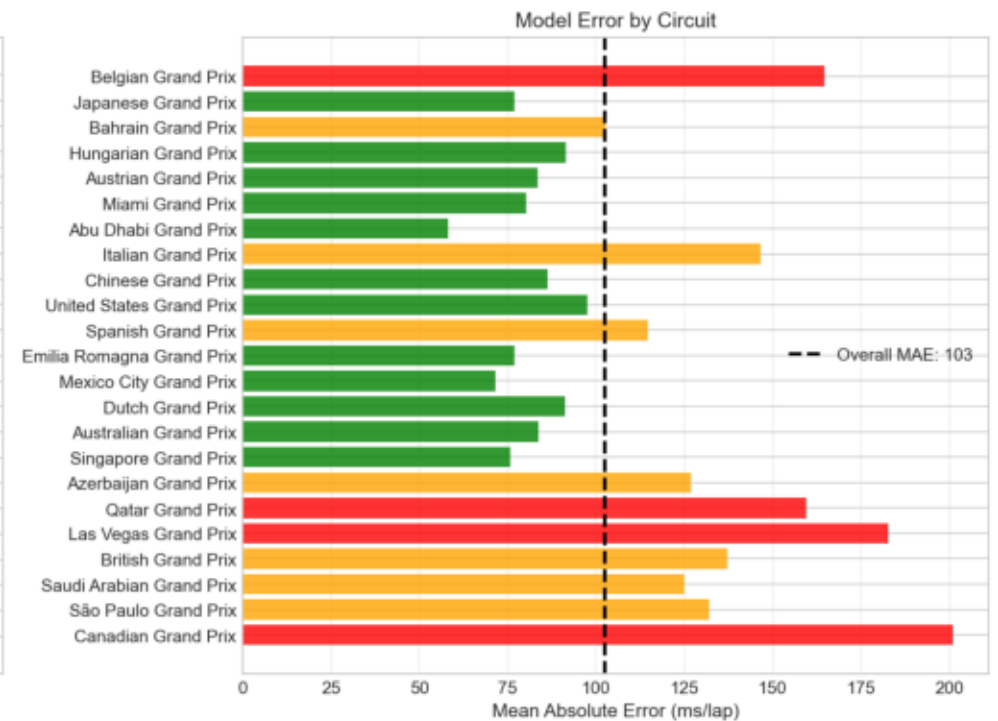
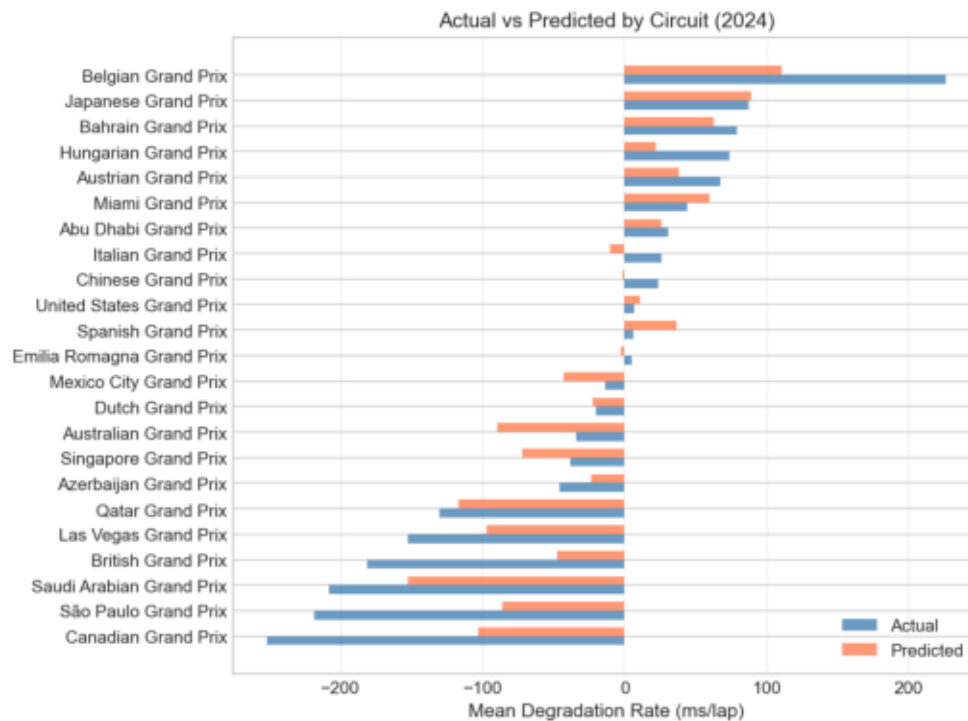
Residual diagnostics (bias, spread, tails, error by regime)



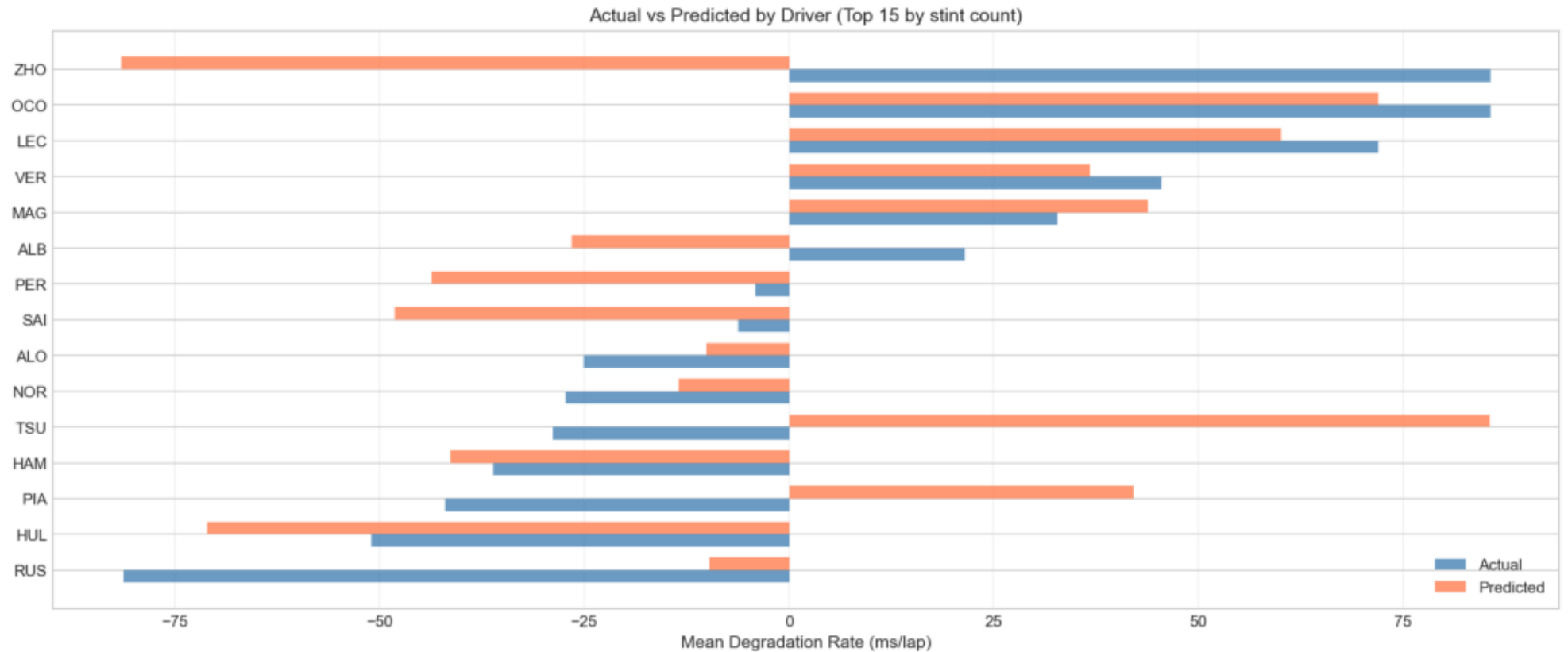
Prediction density (where the model spends its probability mass)



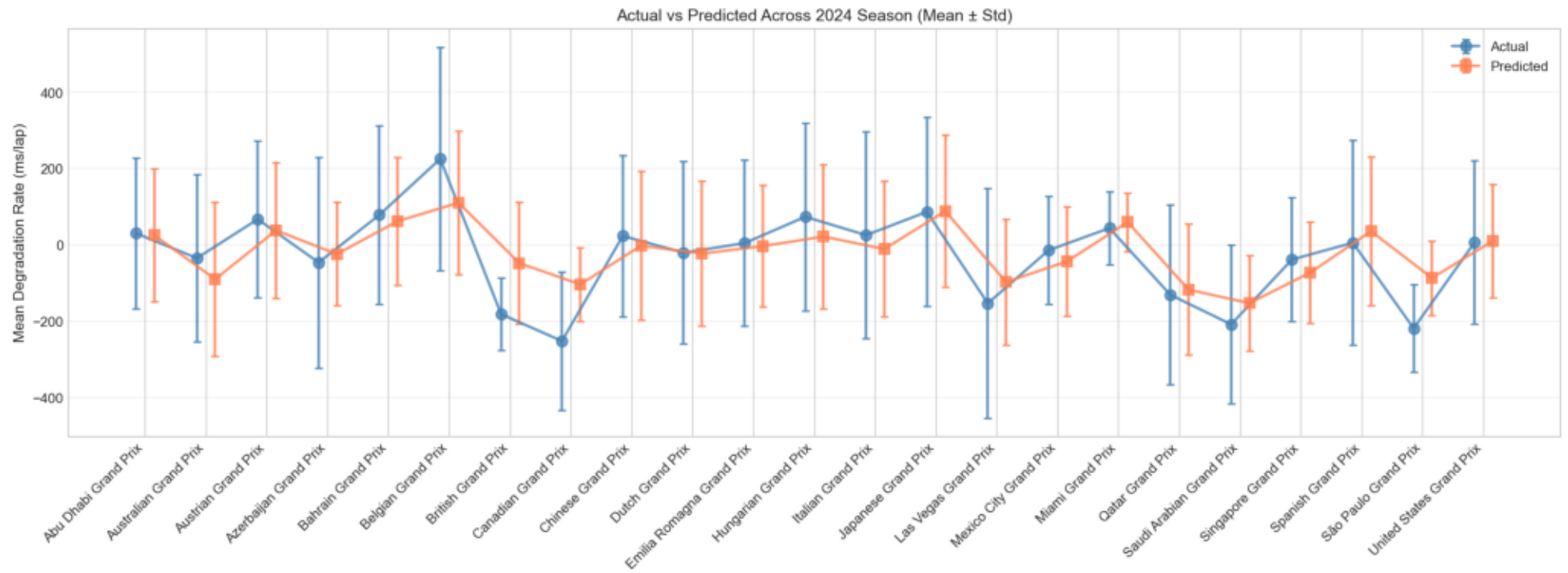
Circuit-level mean fit + error hotspots



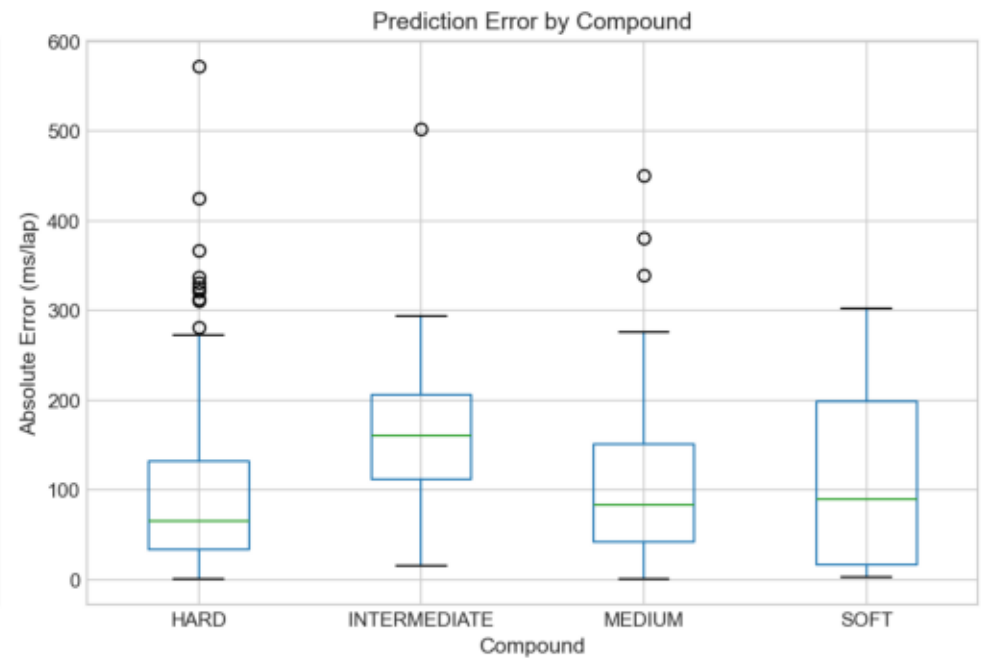
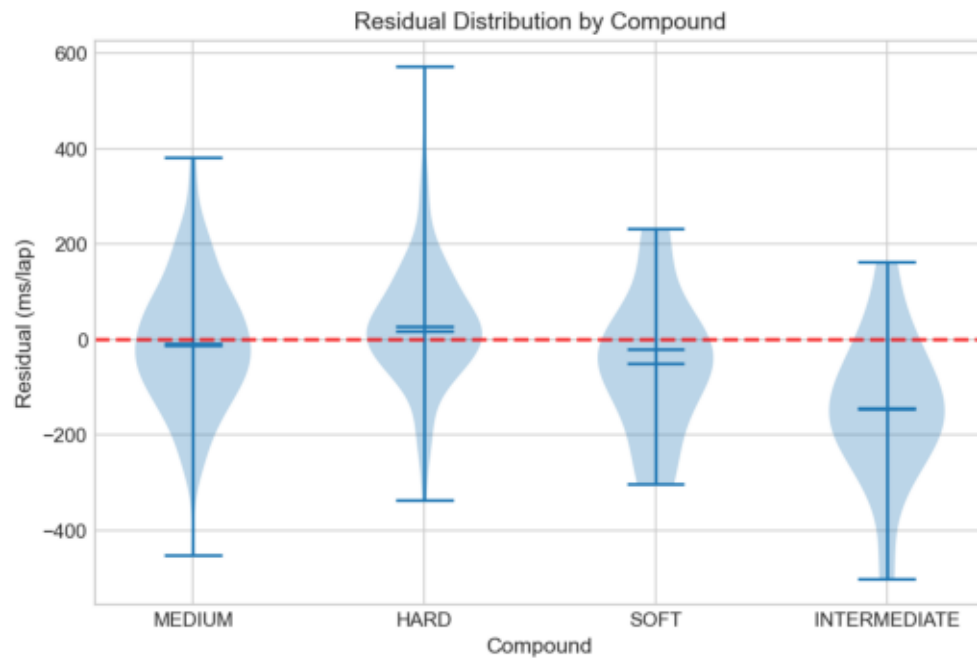
Driver-level mean fit (top drivers by stint count)



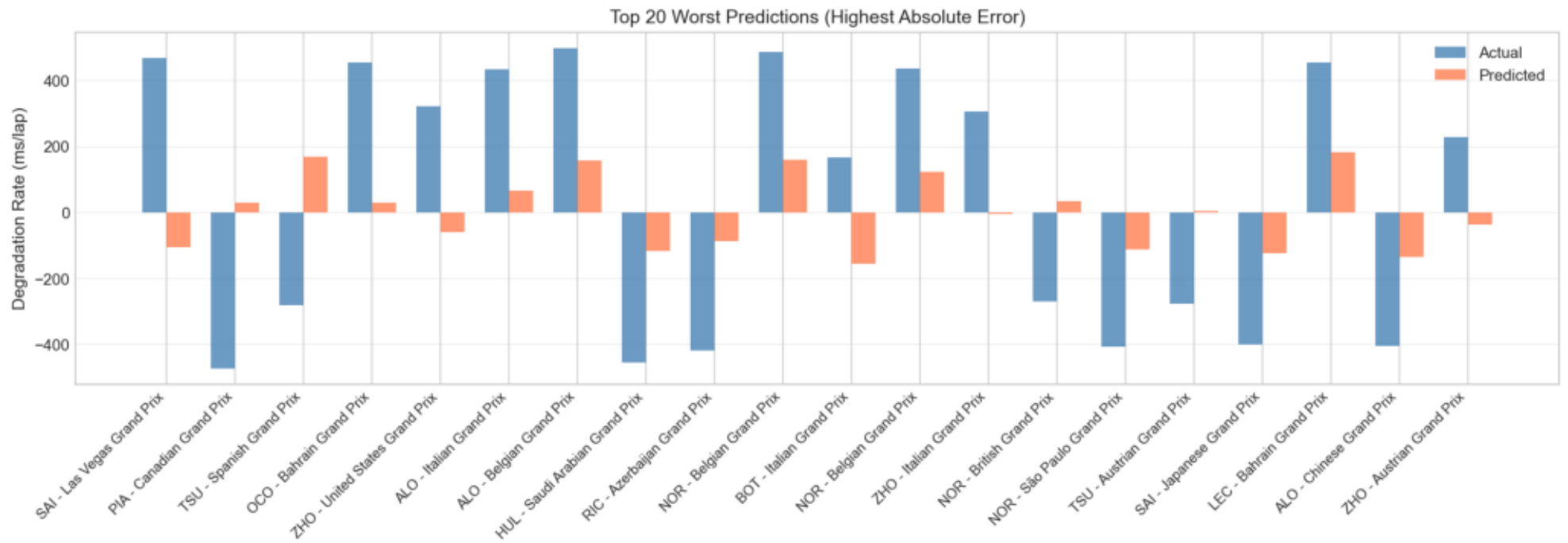
Race-by-race season alignment (mean $\pm$ std)



Compound-specific bias + error distributions



Outliers: the largest misses (where the model breaks)



Next Steps

- Add wet-conditions model (weather + track state features).
- Online calibration: adjust bias using first ~5 laps of a stint.
- Integrate traffic / clean-air features (Domain 2) to capture dirty-air degradation impact.
- Add SHAP-based explanations per prediction for race-engineering interpretability.

Artifacts:

- Figures: C:/Users/USER/Documents/Python Projects/f1-strategy/docs/figures/domain1_modeling
- Report PDF: C:/Users/USER/Documents/Python Projects/f1-strategy/docs/DOMAIN1_TYRE_DEGRADATION_REP